

# Endpoint weak chain upper gradients are not stable under almost-everywhere limits

Autonomous counterexample packet for arXiv:2408.15071

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## Abstract

Caputo–Cavallucci ask whether their almost-everywhere limit stability proposition for weak chain upper gradients remains true for the asymmetric endpoint integrals  $\lambda = 0, 1$ . It does not. On the real line, change the a.e. limit representative at the single endpoint 0. For  $\lambda = 1$ , all one-edge chains  $(x, 0)$  detect that change, and their endpoint modulus is infinite because admissibility forces  $\rho(x) \geq 1/|x|$ . Reversing the construction gives the same negative answer for  $\lambda = 0$ .

## 1 The endpoint question

Fix  $p \geq 1$  and  $\varepsilon > 0$ . For a chain  $\mathbf{c} = (q_0, \dots, q_N)$ , the source defines

$$\int_{\mathbf{c}}^{\lambda} g = \sum_{i=0}^{N-1} (\lambda g(q_i) + (1 - \lambda)g(q_{i+1}))d(q_i, q_{i+1}).$$

Thus  $\lambda = 1$  samples the left endpoint of each edge, while  $\lambda = 0$  samples the right endpoint. A Borel  $g \geq 0$  is a weak  $(\varepsilon, \lambda)$ -upper gradient of  $u$  when

$$u(q_N) - u(q_0) \leq \int_{\mathbf{c}}^{\lambda} g$$

outside a family of chains of  $(\varepsilon, \lambda, p)$ -modulus zero.

Proposition 5.9 proves, for the symmetric integral, that if  $u_j \rightarrow u$  pointwise a.e.,  $g_j$  are weak upper gradients of  $u_j$ , and  $g_j \rightarrow g$  in  $L^p$ , then  $g$  is a weak upper gradient of  $u$ . The source says it does not know whether this remains true at  $\lambda = 1$  (and treats  $\lambda = 0$  as analogous):

case  $\lambda = 0$  being analogous. We have already noticed that we do not know if Proposition 5.9 holds for  $\lambda = 1$ , but in any case  $\text{WUG}_n^{\varepsilon, 1}(u)$  is closed. Moreover it is possible to show,

## 2 A modulus calculation

Let  $X = \mathbb{R}$  with Euclidean distance and Lebesgue measure, and put  $r = \min\{\varepsilon/2, 1\}$ . For  $0 < |x| < r$ , let

$$\mathbf{c}_x^+ = (x, 0), \quad \mathbf{c}_x^- = (0, x).$$

Both are  $\varepsilon$ -chains. Define  $\Gamma_+ = \{\mathbf{c}_x^+ : 0 < |x| < r\}$  and  $\Gamma_- = \{\mathbf{c}_x^- : 0 < |x| < r\}$ .

**Theorem 1.** *For every  $p \geq 1$ ,*

$$\text{Mod}_p^{\varepsilon,1}(\Gamma_+) = \infty, \quad \text{Mod}_p^{\varepsilon,0}(\Gamma_-) = \infty.$$

*Proof.* If  $\rho$  is admissible for  $\Gamma_+$  at  $\lambda = 1$ , then for every  $0 < |x| < r$ ,

$$1 \leq \int_{\mathbf{c}_x^+}^1 \rho = \rho(x)|x|.$$

Hence  $\rho(x) \geq |x|^{-1}$  throughout the punctured interval, and

$$\int_{\mathbb{R}} \rho^p dx \geq 2 \int_0^r x^{-p} dx = \infty.$$

Every admissible function therefore has infinite  $L^p$  cost, proving the first equality. For  $\Gamma_-$  at  $\lambda = 0$  the sampled endpoint is again  $x$ , so  $\int_{\mathbf{c}_x^-}^0 \rho = \rho(x)|x|$  and the same proof applies.  $\square$

### 3 Counterexamples to limit stability

**Theorem 2.** *The endpoint analogues of Proposition 5.9 fail on  $\mathbb{R}$  for every  $p \geq 1$  and every  $\varepsilon > 0$ , both for  $\lambda = 1$  and for  $\lambda = 0$ .*

*Proof.* For  $\lambda = 1$ , take

$$u_j \equiv 0, \quad g_j \equiv 0, \quad u = \mathbf{1}_{\{0\}}, \quad g \equiv 0.$$

Then  $u_j \rightarrow u$  pointwise Lebesgue-a.e. and  $g_j \rightarrow g$  in  $L^p$ . Because every  $u_j$  is constant,  $g_j = 0$  satisfies the endpoint upper gradient inequality on every chain. However, for each  $\mathbf{c}_x^+ = (x, 0) \in \Gamma_+$ ,

$$u(0) - u(x) = 1 > 0 = \int_{\mathbf{c}_x^+}^1 g.$$

The whole family  $\Gamma_+$  is a failure family and has infinite modulus by Theorem 1; it cannot be discarded. Thus  $g \notin \text{WUG}_p^{\varepsilon,1}(u)$ .

For  $\lambda = 0$ , instead take  $u = -\mathbf{1}_{\{0\}}$  while keeping  $u_j = g_j = g = 0$ . Along every  $\mathbf{c}_x^- = (0, x) \in \Gamma_-$ ,

$$u(x) - u(0) = 1 > 0 = \int_{\mathbf{c}_x^-}^0 g.$$

Again Theorem 1 prevents discarding the failure family, so  $g \notin \text{WUG}_p^{\varepsilon,0}(u)$ .  $\square$

### 4 Why the symmetric proof breaks

For  $0 < \lambda < 1$ , chains meeting a measure-zero exceptional set form a modulus-zero family, which makes a.e. convergence sufficient. At  $\lambda = 1$ , the integral ignores the terminal vertex, so a null-set change at the terminal endpoint can be detected by a family of positive or infinite modulus; at  $\lambda = 0$  the same issue occurs at the initial vertex. A repaired endpoint statement would need convergence outside an endpoint-exceptional set of zero endpoint modulus, rather than merely outside a measure-zero set.

## References

- [1] E. Caputo and N. Cavallucci, *Sobolev spaces via chains in metric measure spaces*, arXiv:2408.15071, Proposition 5.9 and the remark on PDF page 22.